Alvaro Guerra

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Education

Texas A&M University — College Station, TX
B.S. Mechanical Engineering — *2021 - 2025*

Experience

Structures Lead — 11-2024 – 06-2025
IGNITORS (Rocketry Team) — *College Station, TX*

- Owned airframe design of 11k-ft Hybrid COTS rocket, led hazardous parachute pyrotechnic testing, ensuring effectiveness of ejection charges in conjunction with altimeter electronics.
- Managed $3.5k BOM and vendor contracts, delivering on schedule and 1% under budget.

Structures and Material Science Designer — 05-2022 – 04-2023
SAE Aero Design Team — *College Station, TX*

- Established test criteria and pass/fail requirements for multiple flight tests with varying amounts of hardware payload.
- Achieved 1st place internationally among 40+ teams by troubleshooting and repairing critical systems minutes before the final competition flight.

Garden Robot Developer — 08-2024 – 05-2025
Garden Cultivation Capstone Project — *College Station, TX*

- Designed fluid and electrical control systems for remote control irrigation, finishing 100% of design goals, all $200 under budget.
- Designed and built Arduino/Pi-driven assistive garden robot electronic schematics; wrote embedded C++ to control drives, pumps, and sensors, boosting water efficiency 26%.

Real Estate Analyst Intern — 05-2021 – 08-2021
River Valley Appraisals of Texas, LLC — *Edinburg, TX*

- Built regression models to refine comparable sales market analysis with Excel on 100+ properties.

Relevant Coursework

MEEN 345 Fluid Mechanics Laboratory (CFD, DAQ) — 11-2023

- Strengthened systems-level thinking by bridging theoretical fluid dynamics with hands-on sensor-based testing and simulation-driven design feedback.
- Validated aerodynamic performance via hands-on testing in the AF1300 subsonic wind tunnel; integrated VDAS software to simulate airflow, define boundary conditions, and collect drag performance data.

MEEN 423 Machine Learning Engineering Data Analysis (Python) — 01-2021 – 04-2021

- Applied ML methods to stress-strain and financial datasets to automate anomaly detection—strengthened data acquisition and interpretation skills essential for integrated test systems.

Technical Skills

Software: SolidWorks, Python *(pandas, NumPy, Matplotlib, BeautifulSoup, PyQt)*, MATLAB, HTML/CSS, Git, Microsoft Office, LabVIEW, VDAS
Testing & Firmware: Sensors (NEO6M, DHT22, NI MyDAQ, Arduino, Raspberry Pi, Wind Tunnel (AF1300), Oscilloscope, Multimeter
Additional: English / Spanish fluency, TRA L1 / L2 (in progress)